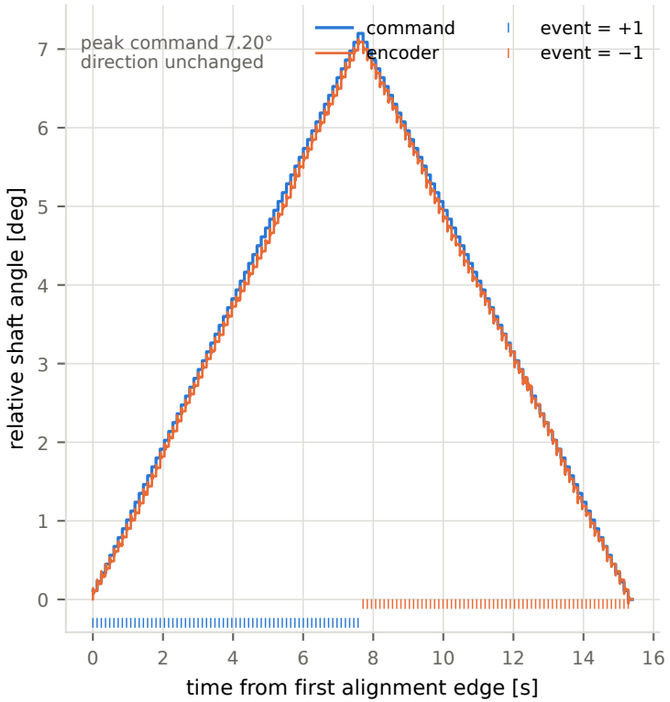
Electrical alignment validation

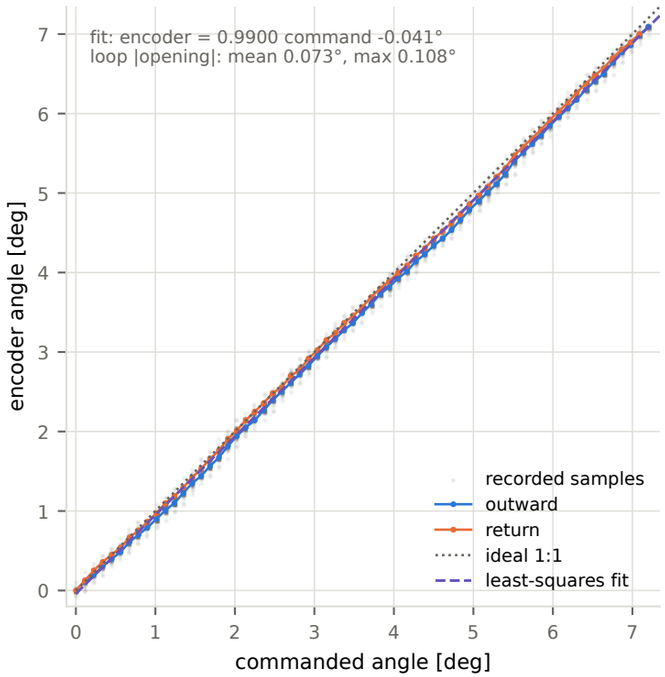
Actions 13–14 · 64 microsteps outward and 64 return · encoder CPR = 20,000

FAIL — 2 of 9 checks

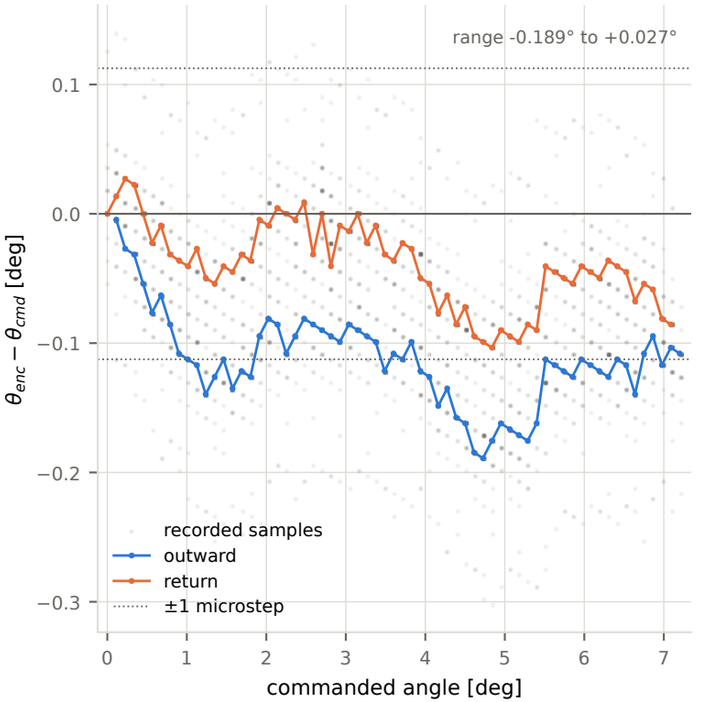
(a) Command and encoder vs time



(b) Encoder vs command



(c) Error vs commanded position



Alignment acceptance table

Settled value = median of final quarter of each 120 ms dwell

Check	Observed	Expected	Status
Mode readback lines	spread=1; intpol=0; IHOLD=IRUN=9	SpreadCycle; intpol=0; IHOLD=IRUN confirmed	PASS
mres column	16	16 throughout	PASS
current_ma	360 mA	close to 360 mA setpoint	PASS
Step events	+1: 64; −1: 64	64 outward; 64 back	PASS
Direction sign	same sign; no flip	encoder and command same sign	PASS
Fitted scale	0.9900 (-0.998%)	1.000 ± ~0.5%	FAIL
Return error	0.000° (0.00 µstep)	< ~1 microstep (0.113°)	PASS
index_count	72 → 72 (Δ+0)	no change, or expected Z change	PASS
Quantisation	6.25 counts/microstep	≥ ~10 counts per microstep	FAIL

Source: motor_identification_run.csv